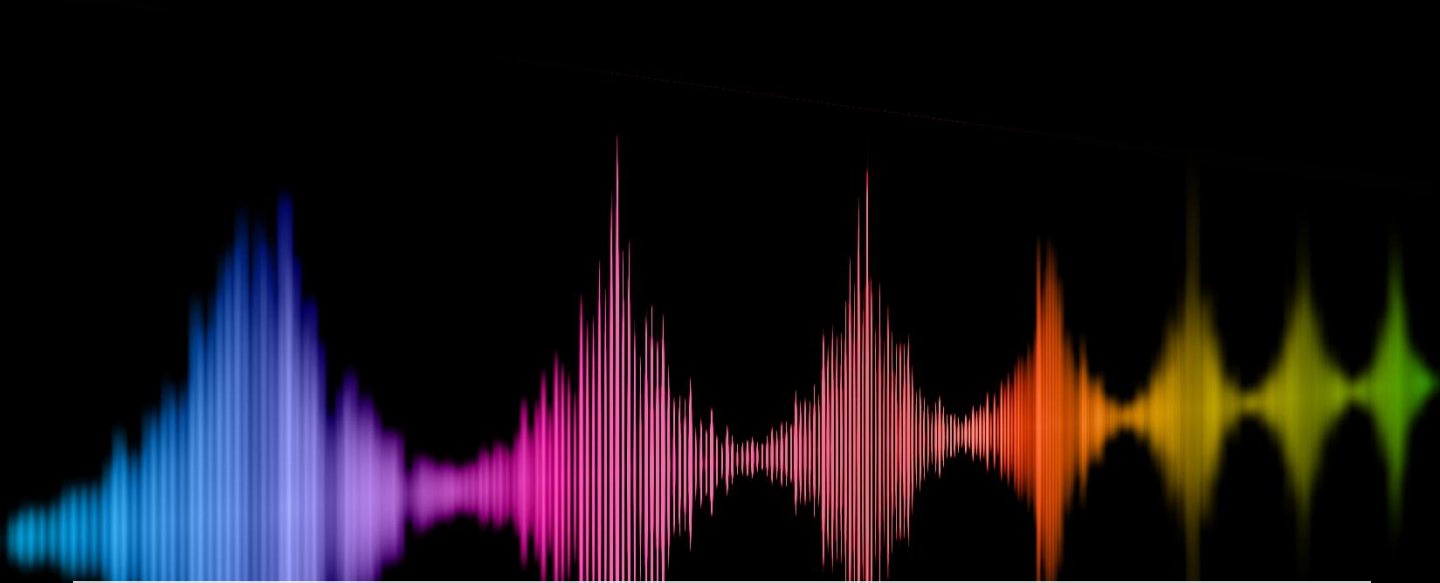




Die wahrgenommene Natürlichkeit von Stimmen

Dr. Christine Nussbaum

Friedrich-Schiller-Universität Jena

Lehrstuhl für Allgemeine Psychologie und kognitive Neurowissenschaften

Wer bin ich



2013 – 2018
2019 – 2023
seit 2023

Psychologiestudium
Promotion
Post-Doktorandin



Dr. Christine Nussbaum

Schwerpunkte: Emotionsverarbeitung in Stimmen
Natürlichkeit in Stimmen

Lehre: Neuropsychologie der auditiven Verarbeitung
Gute Wissenschaftliche Praxis
Empirische Forschungsmethoden

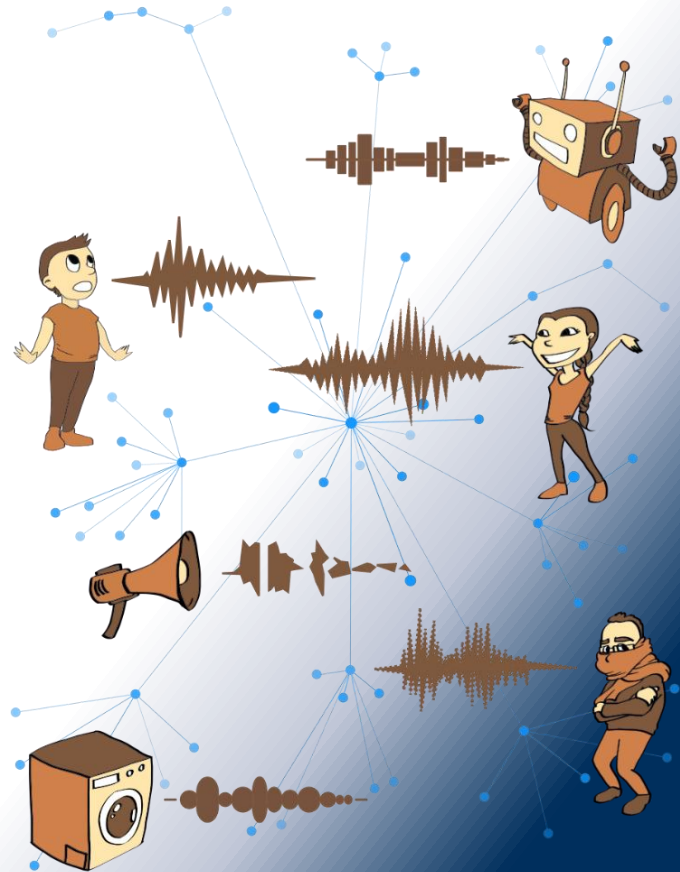
Methoden: Quantitative Forschung
Elektroenzephalographie (EEG)
Voice Morphing

Rollen: Koordinatorin der Jena Voice Research Unit (VRU)
Organisation der „VoicelD 2026“
„VoCSA“ board member



Wenn wir eine Stimmen hören....

.... hören wir eine Person!

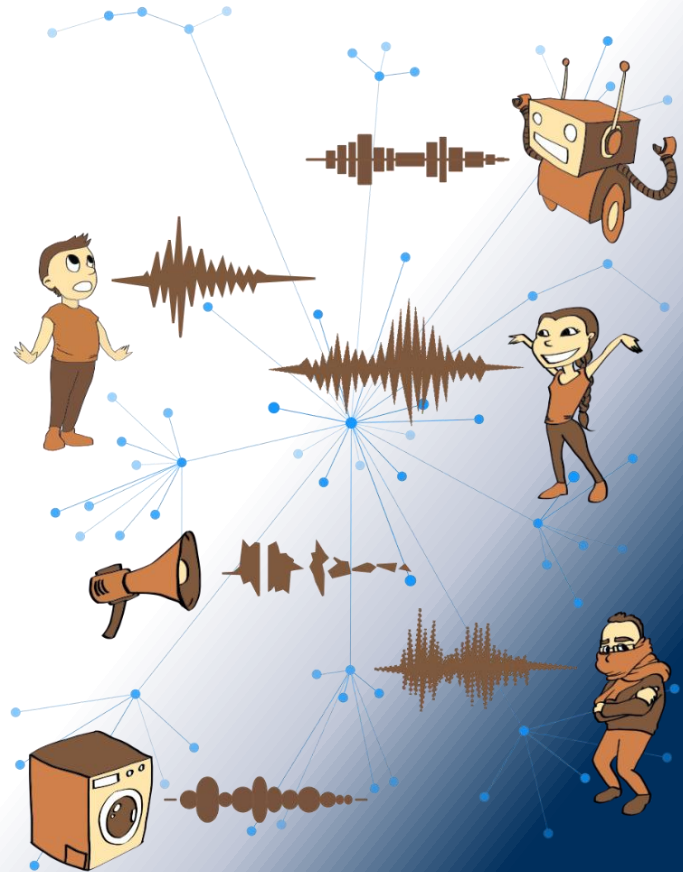


Page 10 of 10



Fig. 2 Word cloud illustrating the multivariate nature of person perception from unfamiliar voices. [Lavan & McGettigan, 2023; Lavan 2023]

**Warum ist es wichtig,
sich mit der Natürlichkeit
von Stimmen zu
beschäftigen?**



Natürlichkeit hat einen (negativen) Einfluss auf die kommunikative Qualität

“Impairments in speech naturalness can lead to communication partners perceiving the affected individuals as unhappy, cold, withdrawn, introverted, or bored. These false perceptions can interrupt participation in regular life roles, leading to loss of employment and independence. Thus, impaired speech naturalness can result in social isolation, reduced quality of life, and depression.” [Stepp & Voijtech 2019]

Natürlichkeit ist ein wichtiger Designaspekt von Sprachsystemen

“The growing popularity of speech interfaces goes hand in hand with the creation of synthetic voices that sound ever more human. Previous research has been inconclusive about whether anthropomorphic design features of machines are more likely to be associated with positive user responses or, conversely, with uncanny experiences. To avoid detrimental effects of synthetic voice design, it is therefore crucial to explore what level of human realism human interactors prefer and whether their evaluations may vary across different domains of application.”

[Schreibelmayer & Mara, 2022]

Natürlichkeit ist wichtig für die Wahrnehmung der eigenen Stimme

Research Articles

A large-scale comparison of two voice synthesis techniques on intelligibility, naturalness, preferences, and attitudes toward voices banked by individuals with amyotrophic lateral sclerosis

Jolene Hyppa-Martin , Jason Lilley , Mo Chen , Jaclyn Friese, Corinne Schmidt & H. Timothy Bunnell 

Pages 31-45 | Received 05 Sep 2022, Accepted 12 Sep 2023, Published online: 04 Oct 2023

 Cite this article  <https://doi.org/10.1080/07434618.2023.2262032>



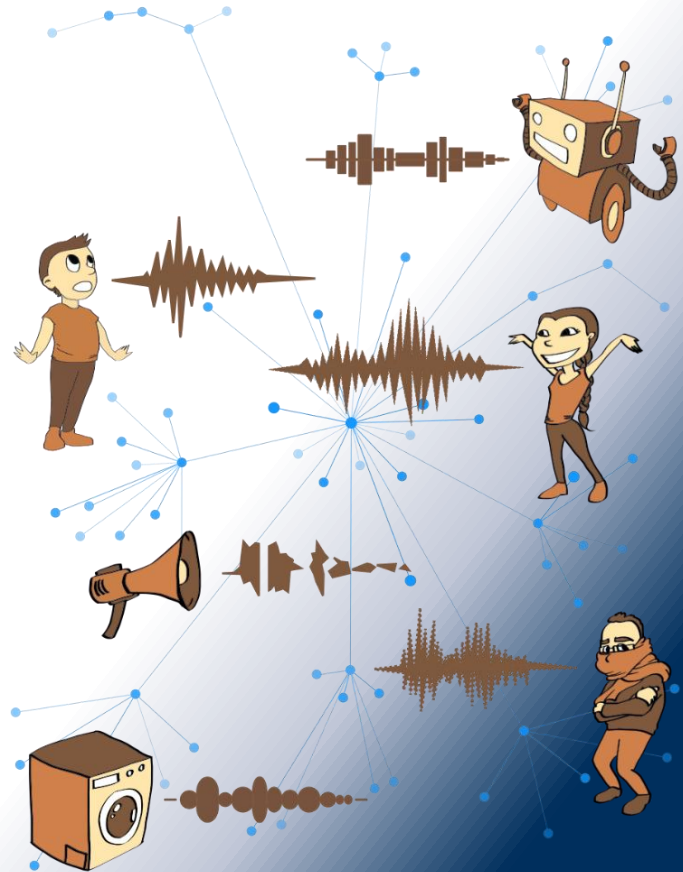
"Identifying natural-sounding synthetic voices for persons who communicate using speech-generating devices is widely regarded as important (Mayo et al., 2011; Nusbaum et al., 1995), but measuring naturalness tends to be less straightforward than measuring intelligibility (Baird et al., 2018; Nusbaum et al., 1995)." (p. 6)

"To examine relative preferences for intelligibility and naturalness, participants were asked whether they would prefer to use a synthetic voice that sounds

- (a) "different from [their] natural voice and is almost always understood,"
- (b) "similar to [their] natural voice and is frequently understood," or
- (c) "exactly like [their] natural voice and is sometimes understood."

Most participants preferred (b), a voice that as similar to their natural voice and frequently understood by others." (p. 16)

Aber was genau **meinen**
wir eigentlich mit
"Natürlichkeit in
Stimmen"?



Was ist eine (un-)natürliche Stimme?

"Naturalness was defined as conforming to the listener's standards of rate, rhythm, intonation, and stress patterning and to the syntactic structure of the utterance being produced"
[Abur et al. 2021, page 4687]

"Naturalness' will not be defined for you." [Martin 1984, page 54]

"Natural speech is the speech most closely perceived as a human voice."
[Mawalim et al. 2022, page 10]

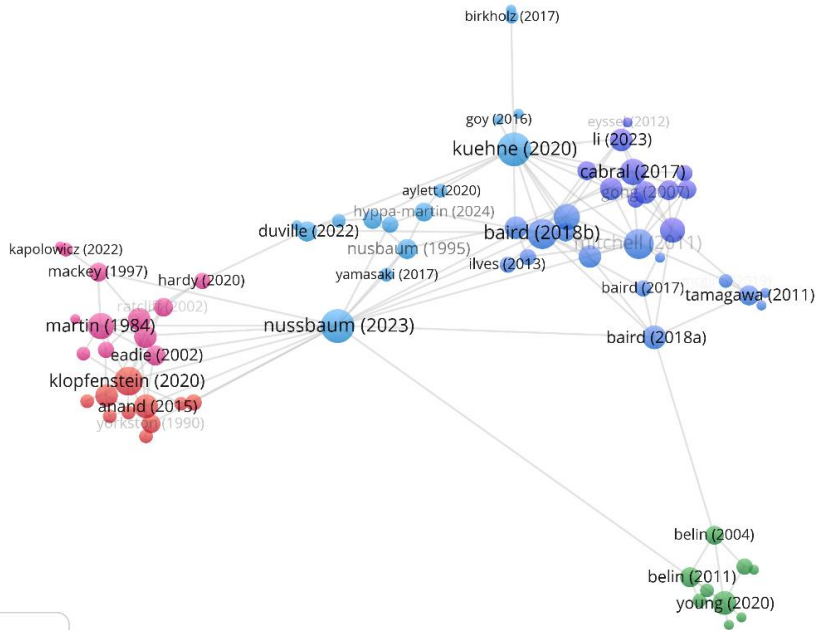
"By naturalness, we understand the voice stimulus to be perceived as a plausible outcome of the human speech production system."
[Nussbaum, et. al. 2023, page 1]

- <50% der wiss. Publikationen haben überhaupt eine explizite Definition

Viele Begriffe für verwandte Konzepte – Synonyme von “Natürlichkeit”

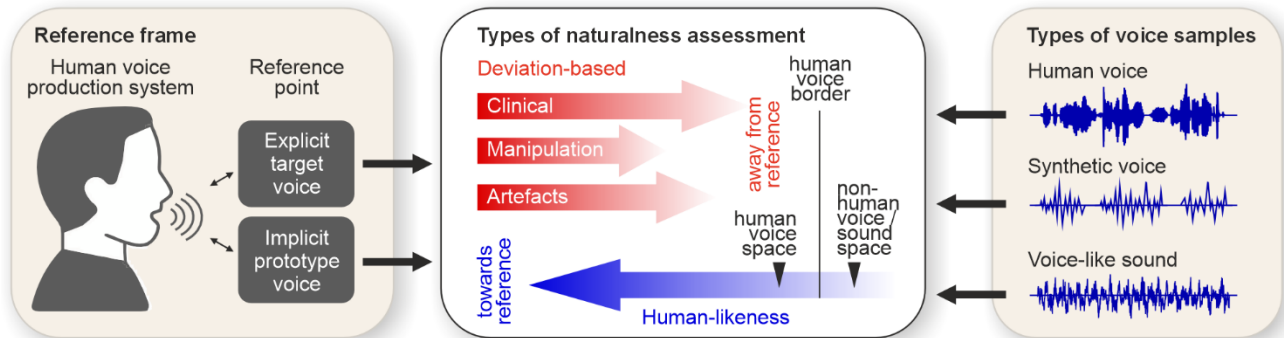


Natürlichkeit in Stimmen – eher ein Flickenteppich als ein Forschungsfeld



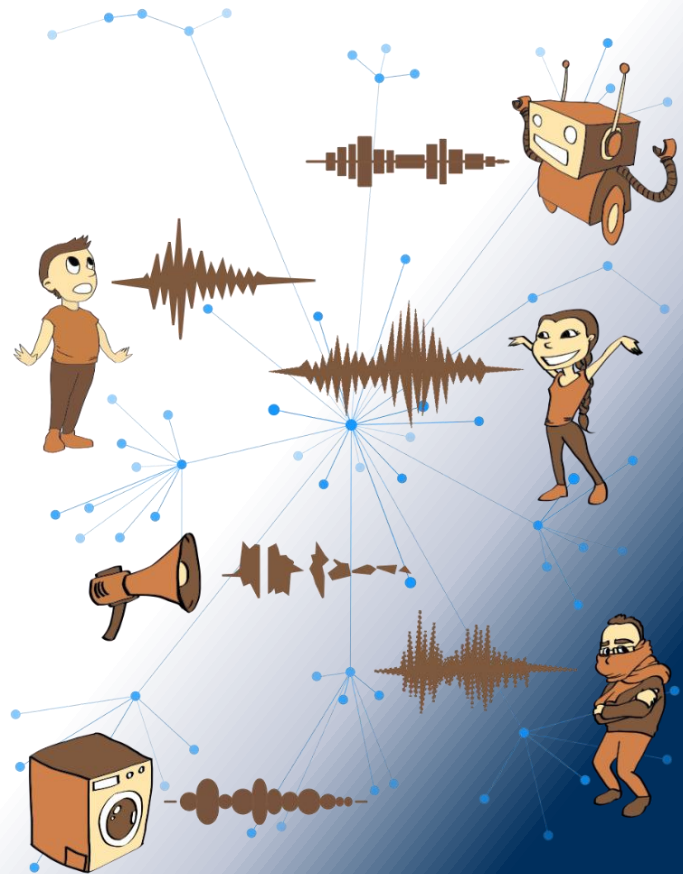
Understanding Voice Naturalness (2025)

Christine Nussbaum, Sascha Frühholz and Stefan R. Schweinberger
In *Trends in Cognitive Sciences*



A conceptual framework for the definition of voice naturalness

Und wie können wir die
Natürlichkeit von
Stimmen **jetzt messen?**



Messung von Natürlichkeit

➤ subjektives Erleben erfragen

Wie natürlich klingt diese Stimme für Sie?

Klingt die Stimme menschlich oder synthetische?

Warum klingt diese Stimme unnatürlich?

Klingt diese Stimme verzerrt oder robotisch?

Welche der beiden Stimmen klingt natürlicher?

Wie menschen-ähnlich klingt diese Stimme?

Welche (akustische) Konfiguration ist natürlicher?

"It sounds like my toaster is speaking to me."
[Kühne et al. 2020]

Messung von Natürlichkeit

Klang diese Stimme natürlich oder unnatürlich?

Natürlich

Unnatürlich

Drücken Sie
„F“

Drücken Sie
„J“

Klang diese Stimme menschlich oder synthetisch?

Menschlich

Synthetisch

Drücken Sie
„F“

Drücken Sie
„J“

Wie natürlich klang diese Stimme?

Natürlich

Unnatürlich



Aufgabenbeschreibung Teil 1

Im Folgenden werden Ihnen verschiedene **Tonaufnahmen** menschlicher Stimmen vorgespielt.

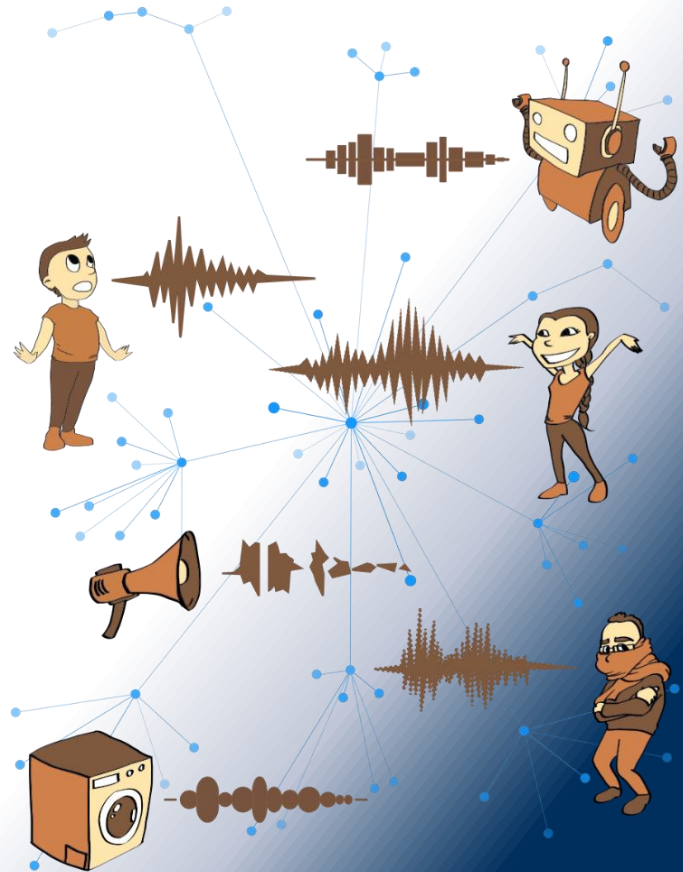
Bei jeder Tonaufnahme handelt es sich um ein **Pseudowort**, also kein tatsächlich im deutschen Sprachraum existierendes Wort. Die Stimmen klingen zum Teil sehr emotional.

Ihre Aufgabe besteht darin, zu entscheiden wie **menschenähnlich** sich die Stimmen anhören.

→ Mit **menschenähnlich** ist in diesem Kontext gemeint, dass die Stimmen sich tatsächlich menschlich/natürlich anhören und nicht auf irgendeine Art verzerrt oder robotisch klingen.

* Weiter durch Drücken der Leertaste *

Und was ist mit der
Wahrnehmung der
eigenen Stimme?



From Voice to Self: An Integrative Framework on Self-Voice Processing

Pavo Orepic and Ana P. Pinheiro
In Perspectives on Psychological Science

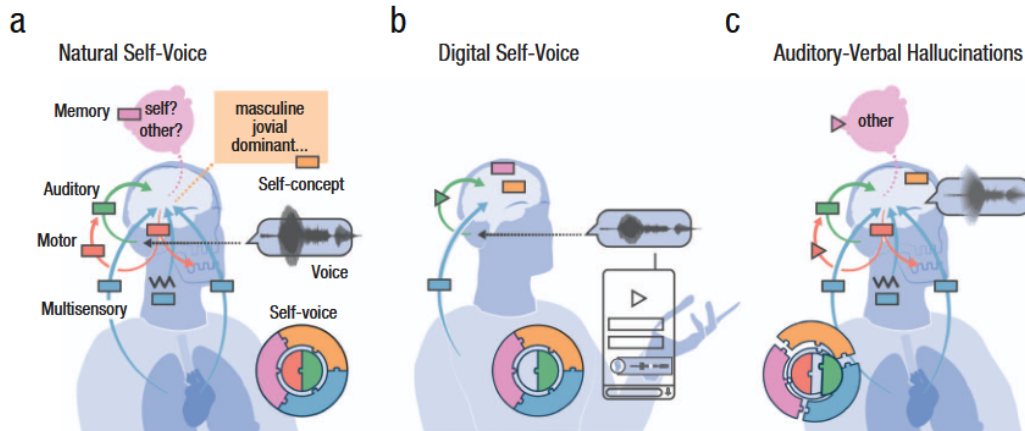


Fig. 1. Self-voice building blocks. In (a) natural self-voice perception (during speaking), all five blocks combine to form a complete self-voice representation (puzzle inset). The auditory block (green) processes the acoustic signal (voice waveform). The motor control block (red) generates the voice (downward arrow) and sends sensorimotor predictions to the auditory block (upward arrow) to stabilize perception. The memory block (pink) provides an internal self-voice template for recognizing whether a voice is self-generated or externally generated. The multisensory integration block (blue) incorporates exteroceptive (e.g., bone-conducted vibrations) and interoceptive cues (e.g., respiration). The self-concept block (orange) aligns the perceived voice with personality and social identity. The lack of motor control involvement (incomplete puzzle) and bone-conducted cues in (b) digital self-voices alters perceived voice quality (changed waveform; triangle in the auditory block). In (c) auditory-verbal hallucinations, the blocks fail to integrate properly (distorted puzzle), producing a hallucinated voice (blurry waveform). Disruptions in sensorimotor prediction (triangle in red upward arrow) and in the memory block (pink triangle) contribute to self-other voice confusion.

Besonderheiten bei Natürlichkeit der eigenen Stimme

Akustische
Komponente

Motorische
Komponente

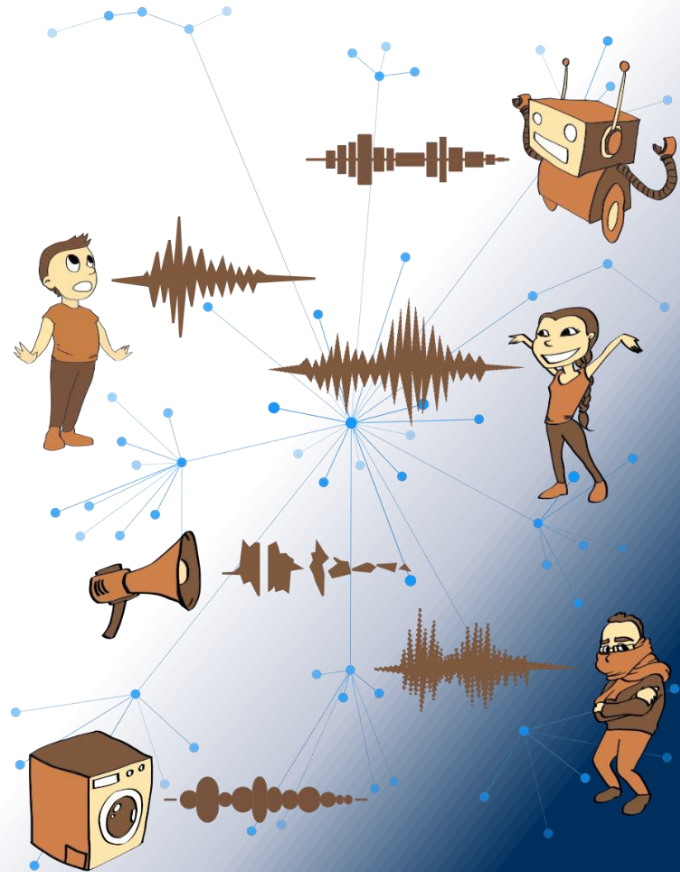
Multisensorische
Komponente

Motivationale/emotionale
Komponente

Gedächtnis-
komponente

Außenwirkungs-
komponente

Offene Fragen in der Forschung zu Natürlichkeit von Stimmen



Offene Fragen

Ibrahim Vefa Arslan, PhD-student
on the project „*Neurocognitive
processing of voice naturalness in
human and synthetic voices*“,
funded by the EU-MSCA Network
„Voice Communication Sciences“

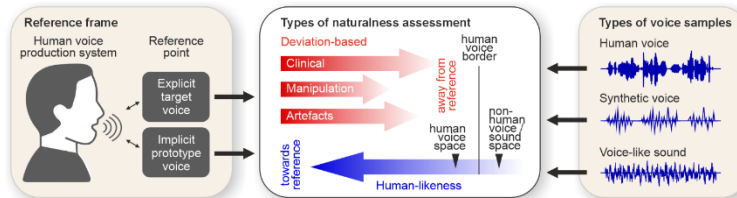


Verarbeitung
im Gehirn

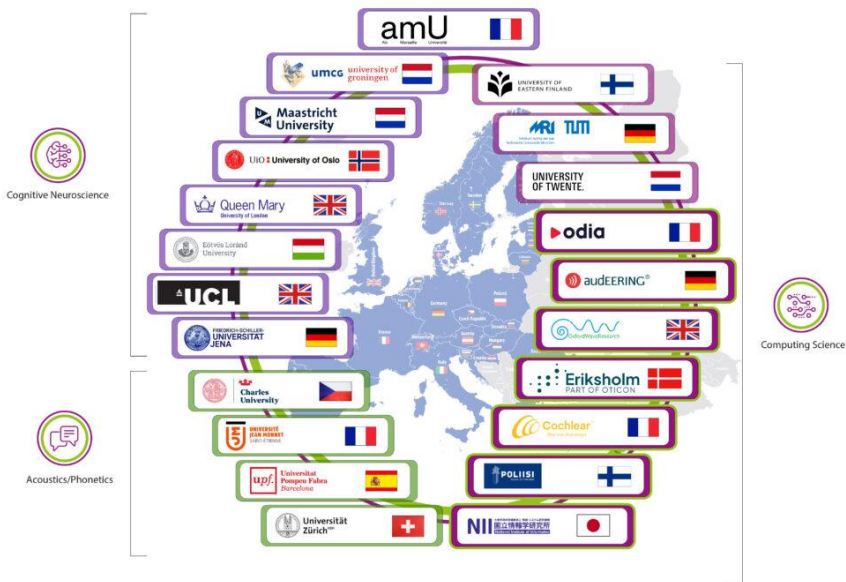
Kontext-
abhängigkeit

Individuelle
Unterschiede

Flexibilität/
Anpassung

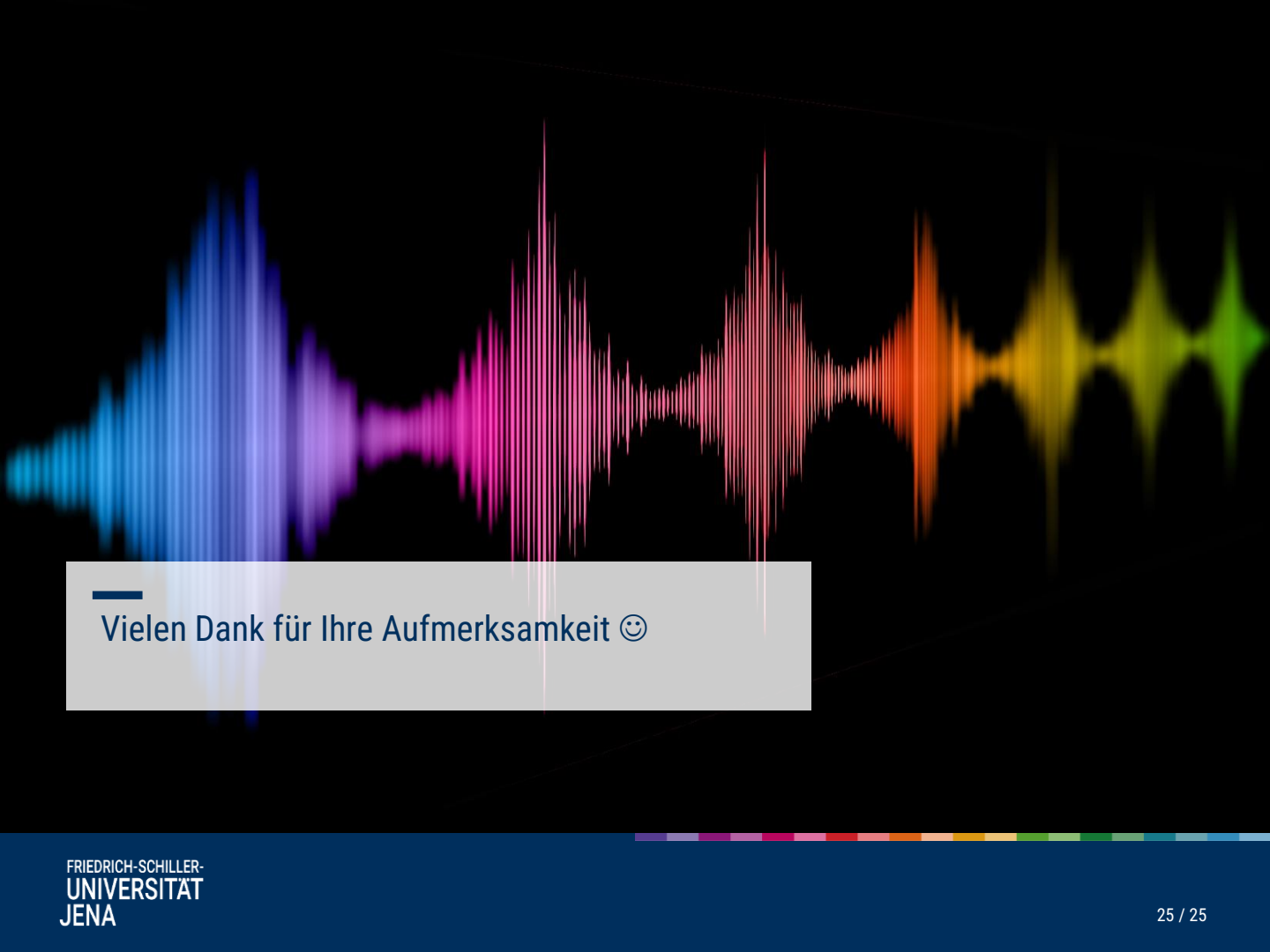


Voice Communication Sciences (VoCS) in Europa





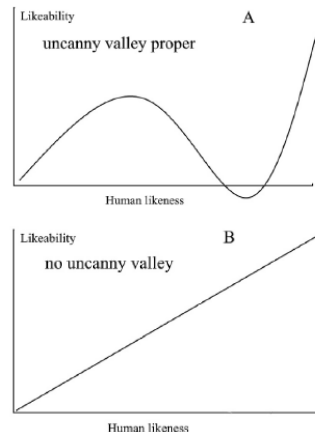
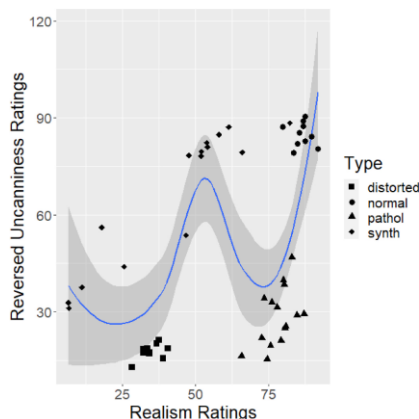
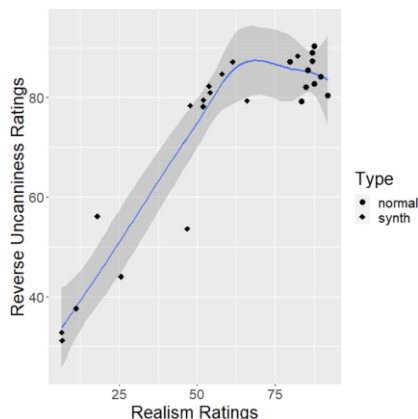
02.-04. September 2026
Jena - Germany



—
Vielen Dank für Ihre Aufmerksamkeit 😊



Deviation from typical organic voices best explains a vocal uncanny valley

Alexander Diel ^{a,b,c,*}, Michael Lewis ^a^a Cardiff University School of Psychology, Park Pl 70, Cardiff, CF10 3AS, UK^b University of Duisburg-Essen, Clinic for Psychosomatic Medicine and Psychotherapy, LVR University Hospital Essen, 45147, Essen, Germany^c Center for Translational Neuro- and Behavioral Sciences (C-TNBS), University of Duisburg-Essen, 45147, Essen, Germany

Understanding voice perception

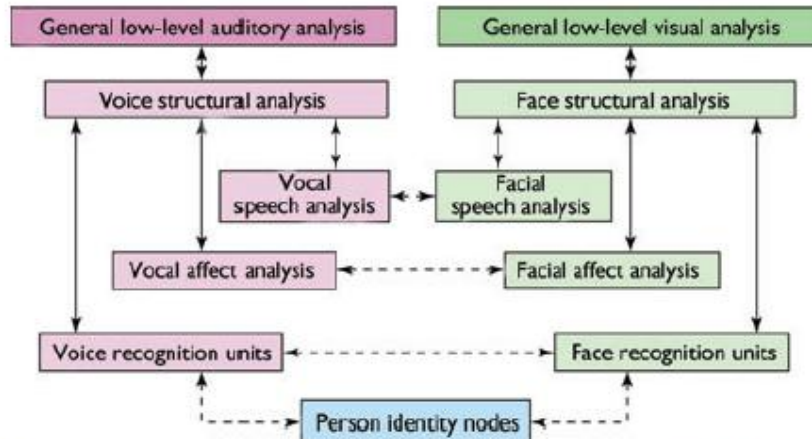


Figure 1. A model of voice perception. Reproduced from Belin *et al.* (2004). After a stage of voice structural encoding restricted to vocal sounds, three partially dissociable functional pathways are proposed to process the three main types of vocal information: speech, identity, and affect. These pathways are analogous to and interacting with equivalent functional pathways involved in facial processing.

Belin *et al* (2011)

Person Perception from Voices (PPV)

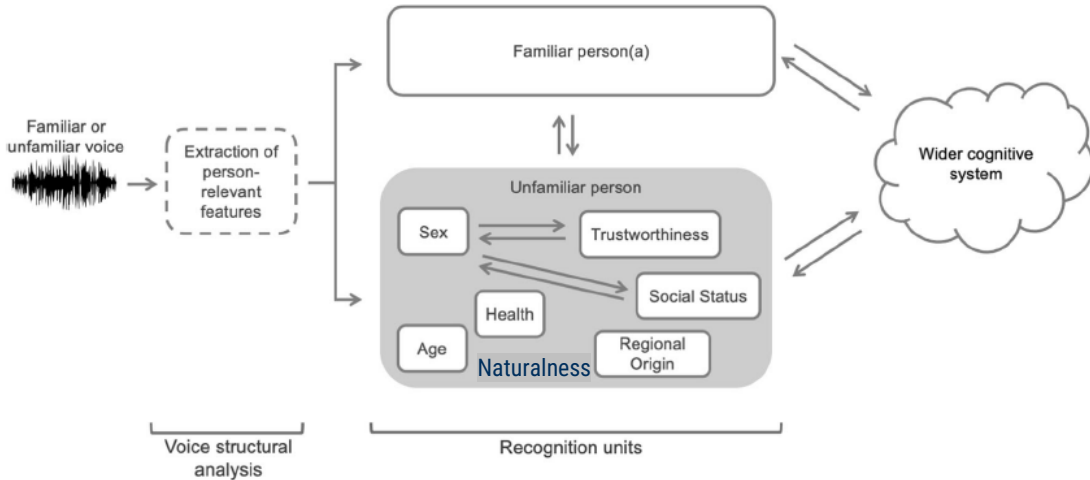


Fig. 3 A hierarchical model of person perception from voices (PPV). The model illustrates how different person characteristics may be recognised and may interact and inform each other during person perception from voices. Boxes with a solid outline describe a recognised percept and the box with a dashed outline describes a computational process. Brackets at the bottom of the figure loosely map the processing stages of the PPV to Belin et al.'s hierarchical model of voice identity perception¹⁰. Not all possible interactions between percepts are shown.

(Lavan & McGettigan 2023)